



POLYTECH.MONS

Audio Processing

How is speech processed in a cell phone conversation?

Every cell phone solves 10 linear equations in 10 unknowns every 10 milliseconds.

(in fact, every 20 ms)

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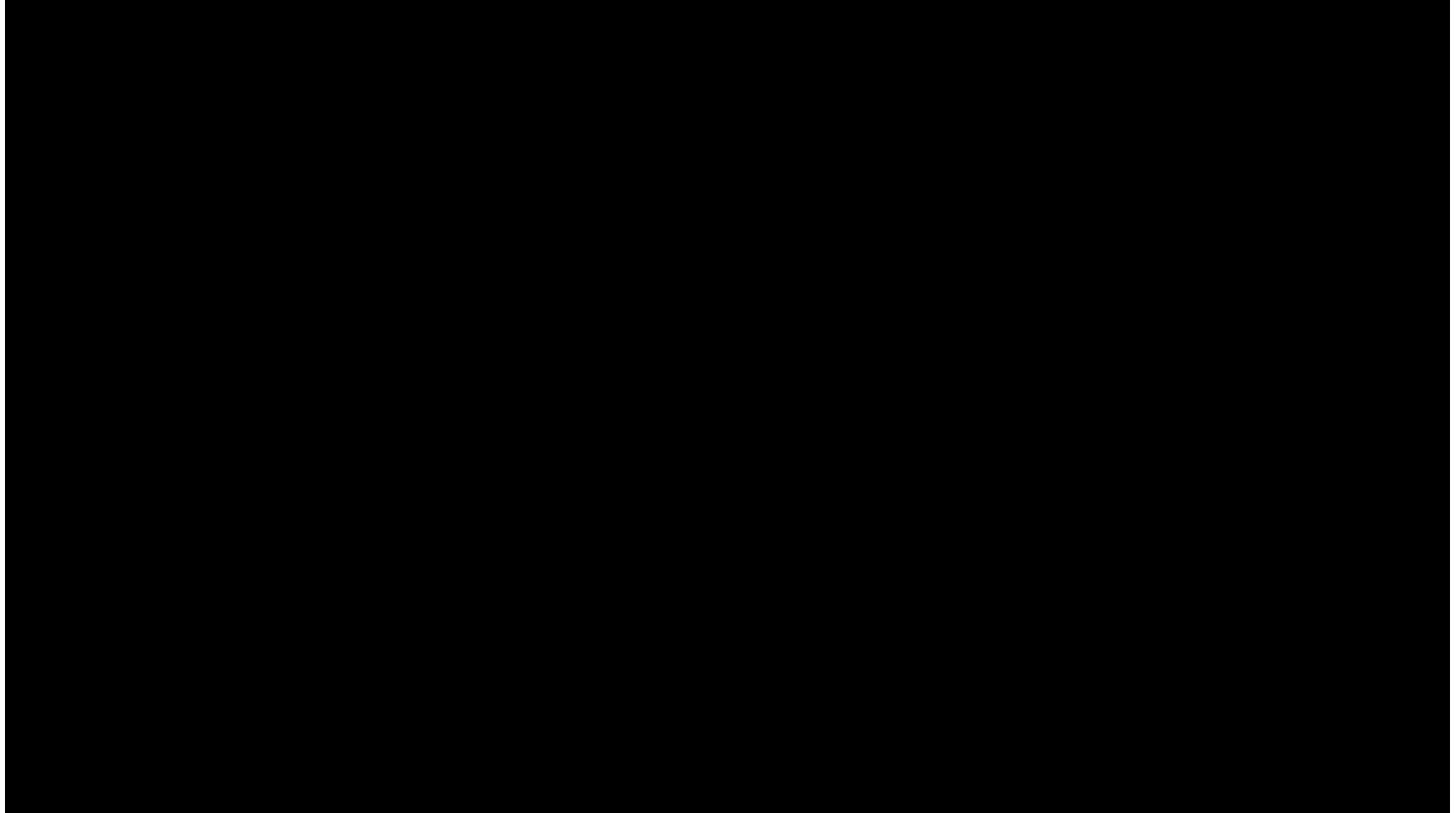
(°) Faculté Polytechnique de Mons, Belgium

(*) Ecole Nat. Sup. des Télécoms, Paris, France

(+) LSI, Allentown, PA, USA

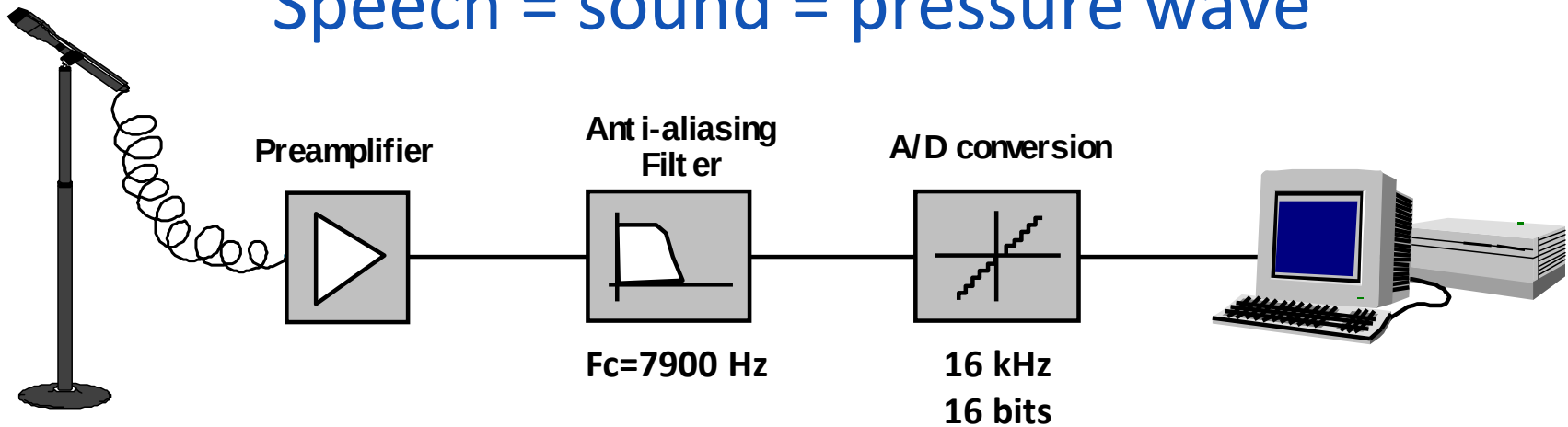


The history of the vocoder



1. Speech acoustics

Speech = sound = pressure wave



Acoustic features

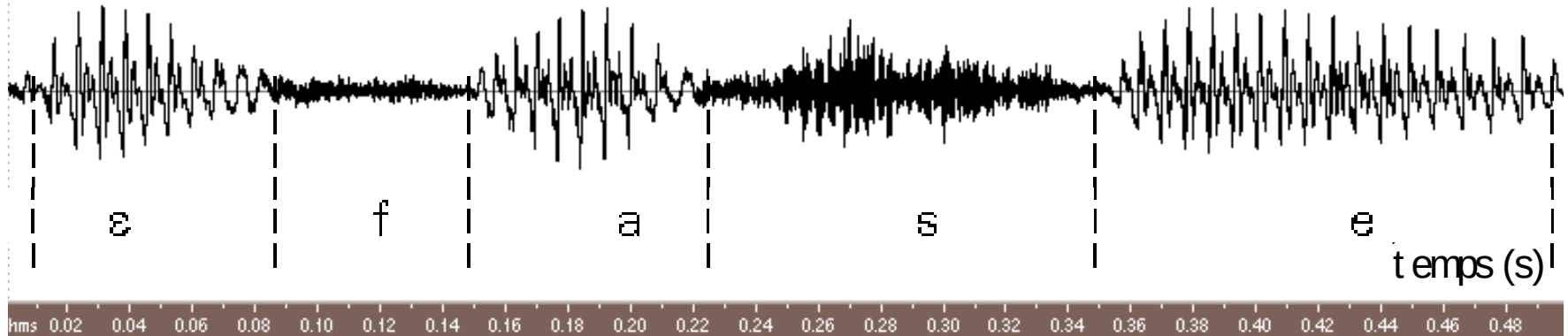
Fundamental frequency, F_0 , Pitch

Amplitude, loudness

Spectrum, timber

The speech waveform

« effacer »



Q : Which acoustic features do you see?

Voiced

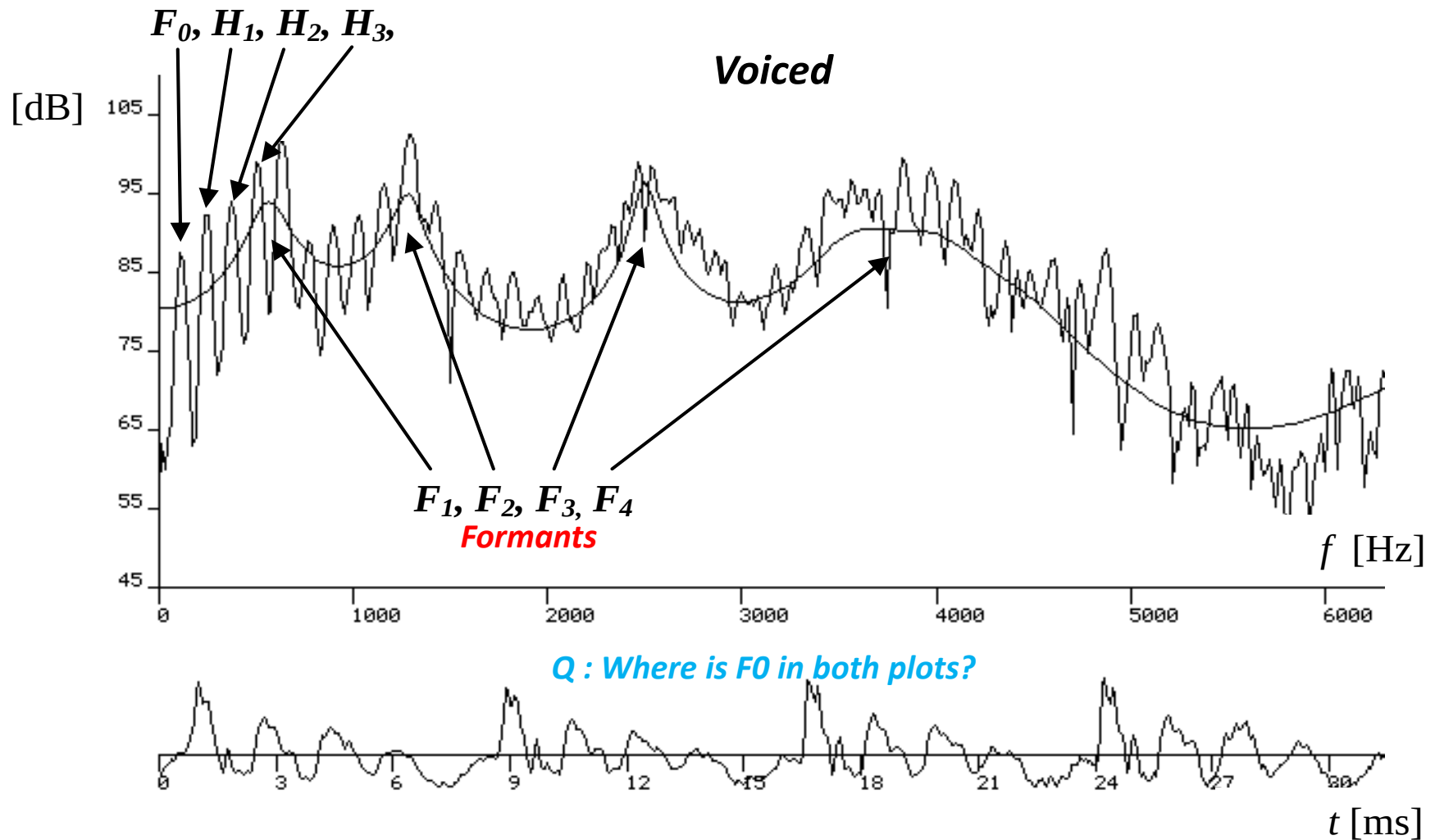


Amplitude, $F_0=1/T_0$

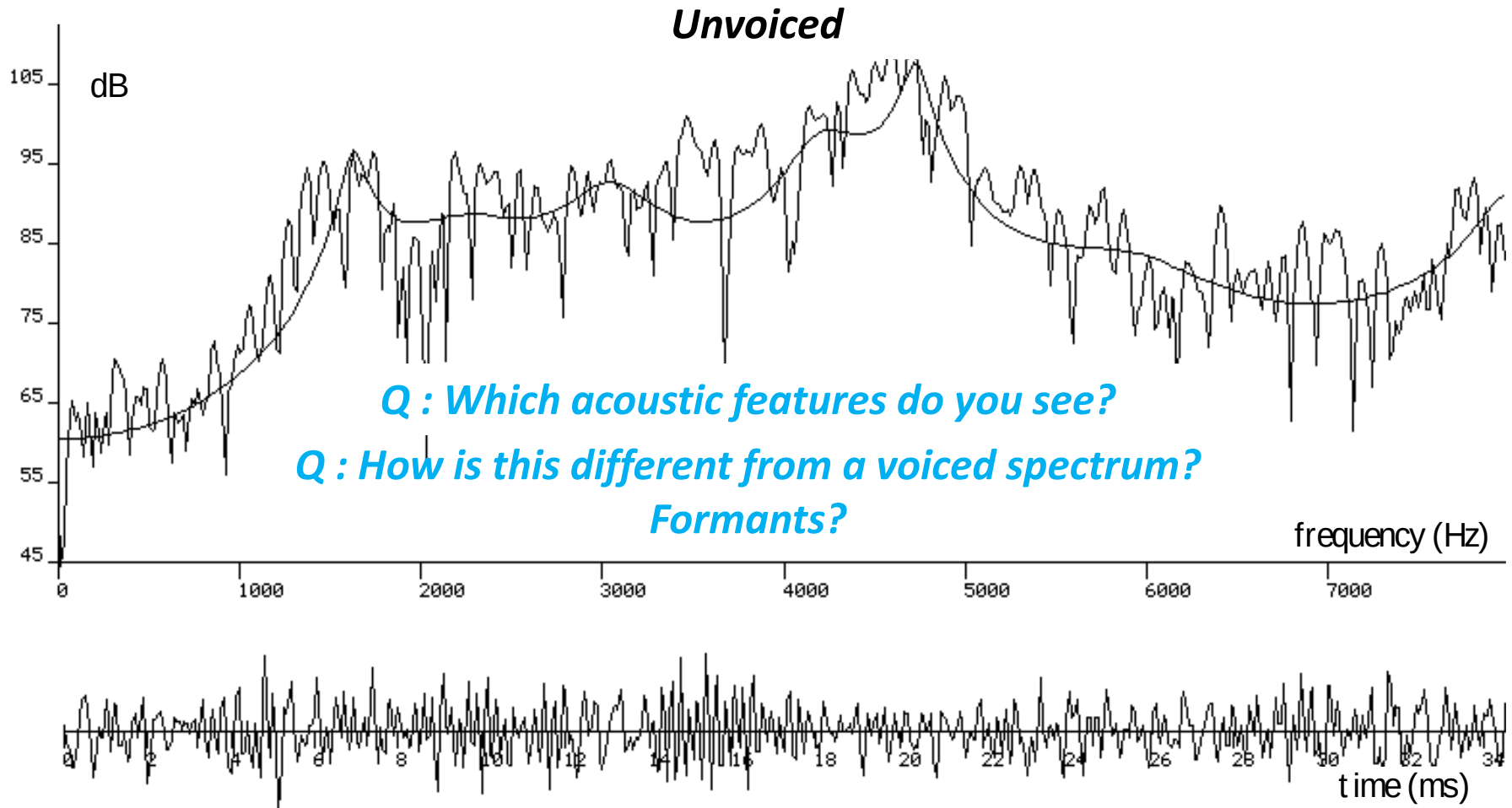
Unvoiced



The speech spectrum

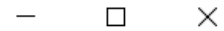


The speech spectrum

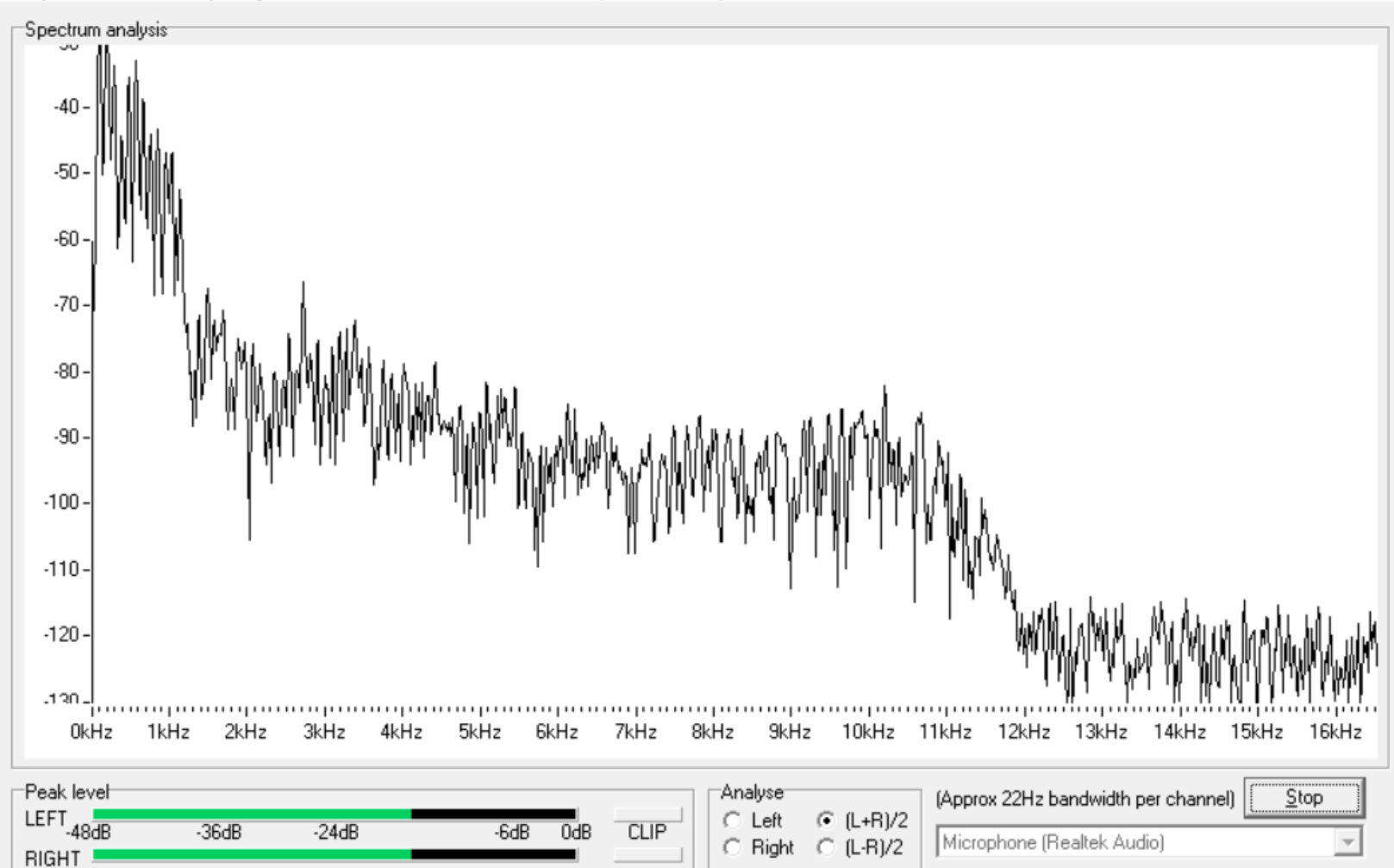


Test?

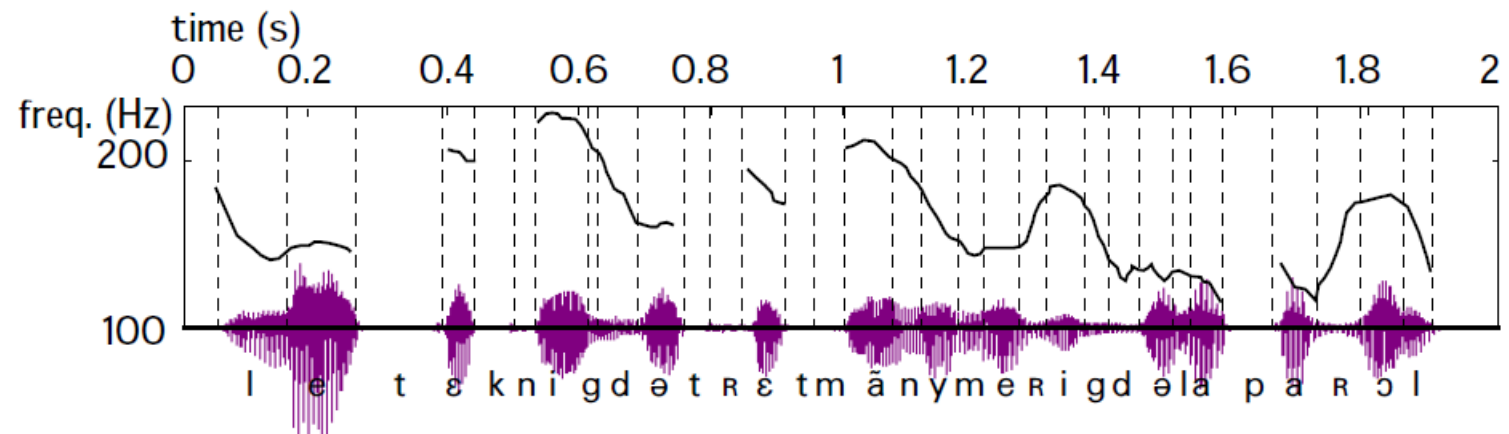
Spectrum analysis v3.5 ©W.A.Steer 2001-2013



Amplitude scale Frequency scale Mixer... Visualisation Sample rate Help



Pitch analysis

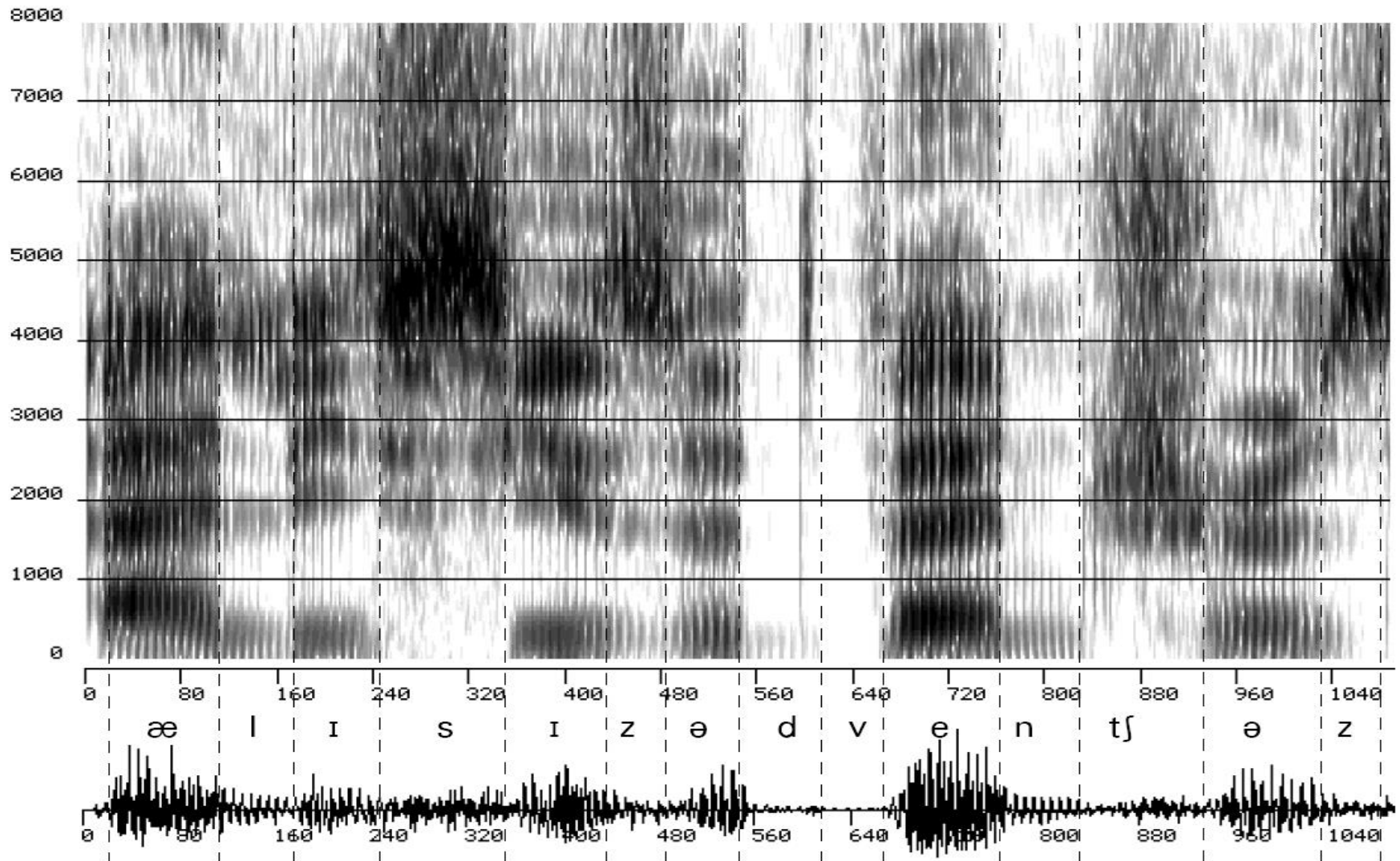


men : 70-250 Hz

women : 150-400 Hz

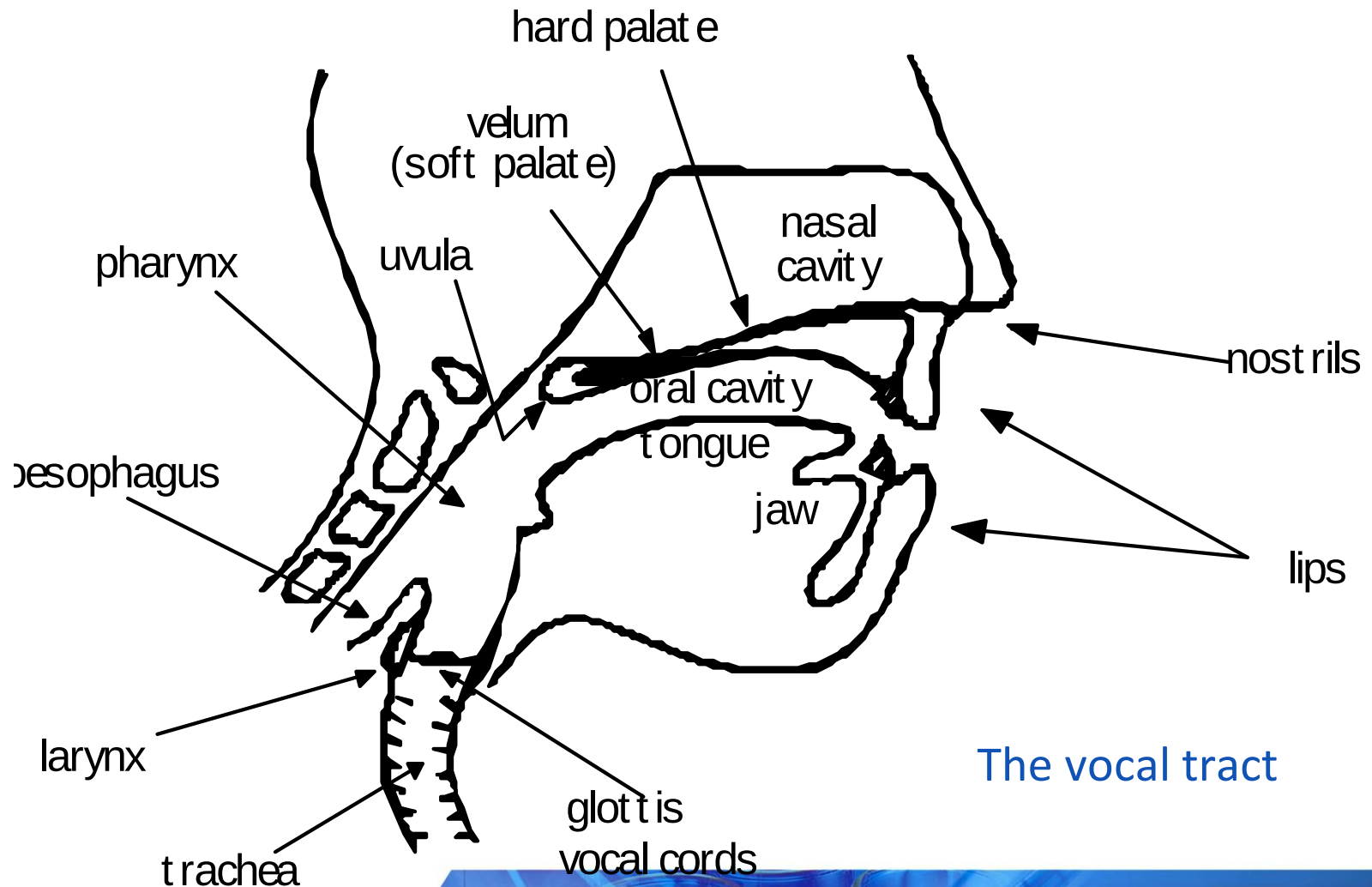
kids : 200-600 Hz

The speech spectrogram

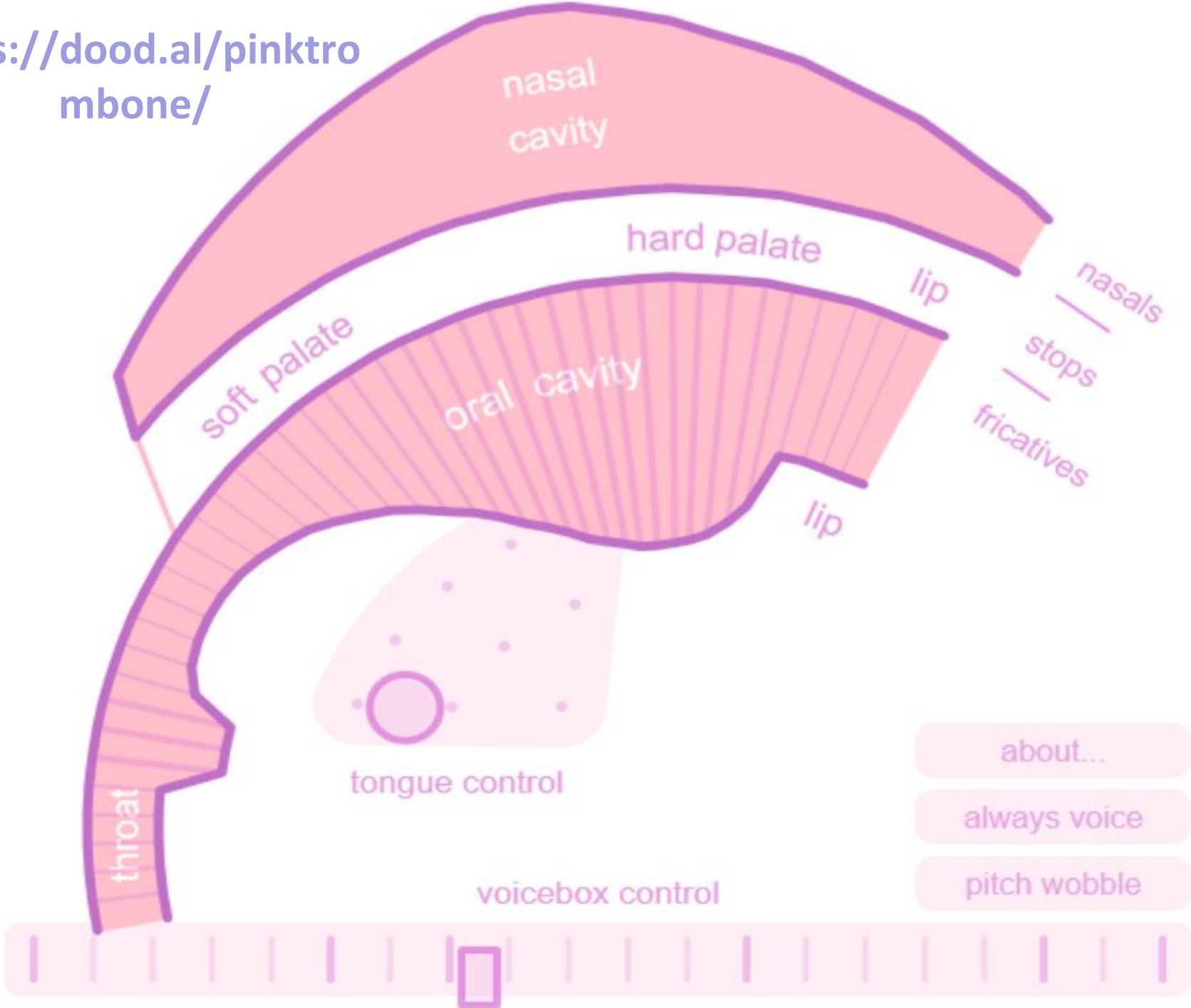


Q : Which acoustic features do you see?

Articulatory phonetics

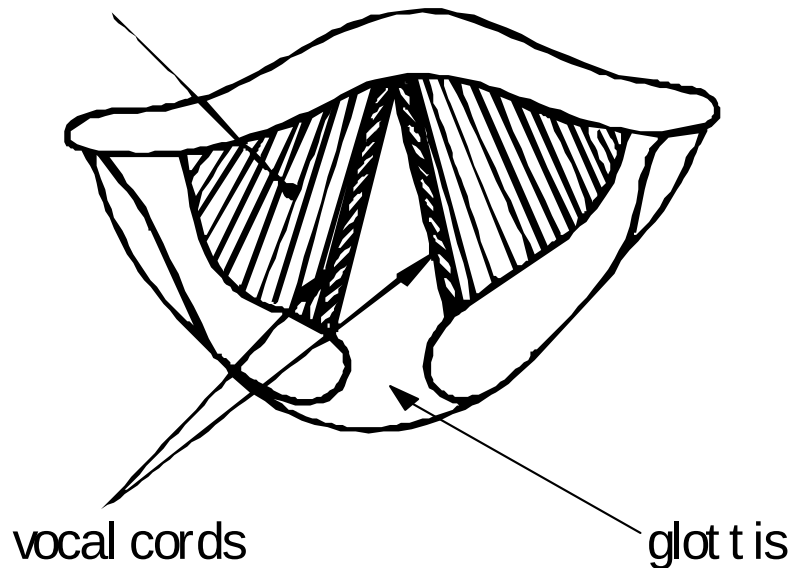


<https://dood.al/pinktro mbone/>



Articulatory phonetics

mucous membrane



The vocal folds

- ❑ Membrane tensed by the muscles
- ❑ Pressure build-up from lungs
- ❑ Opening due to pressure
- ❑ Pressure release
- ❑ Closing



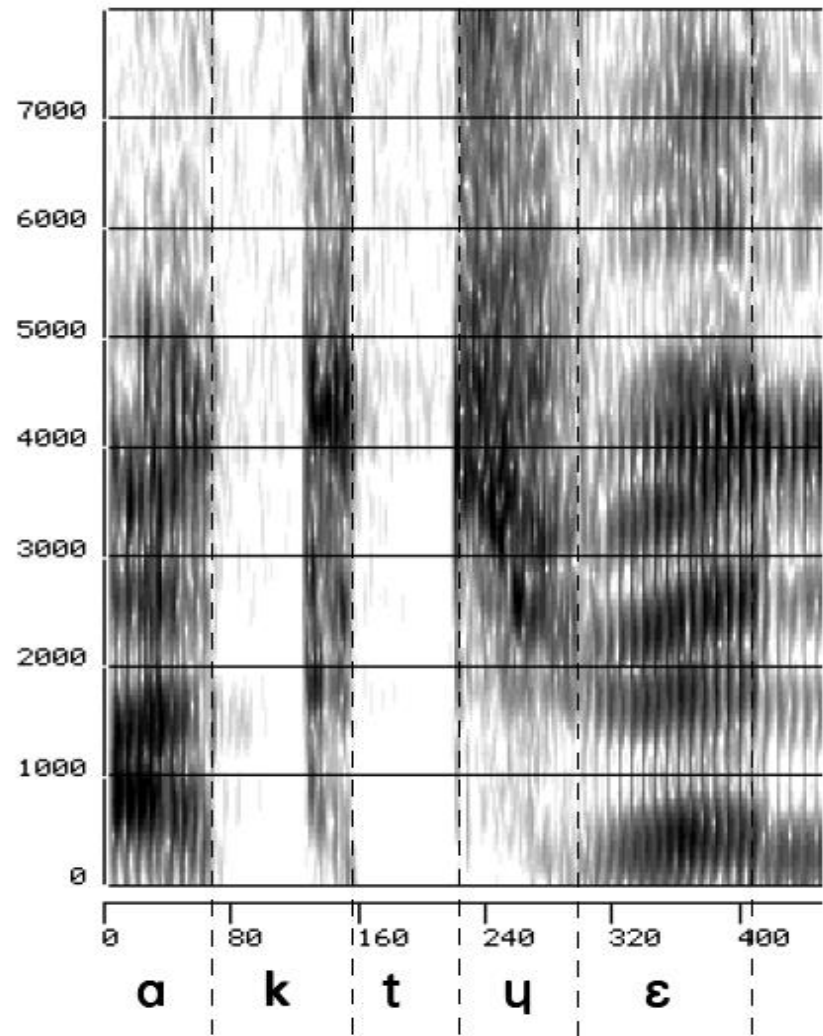
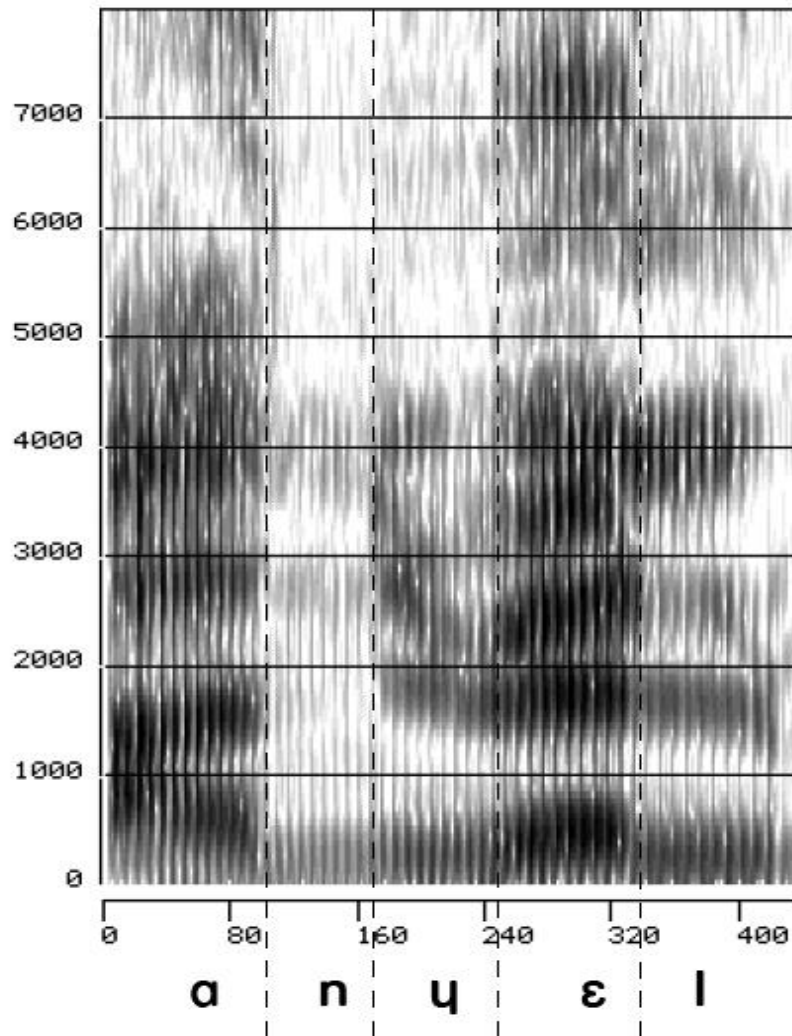
Phonemes

- ❑ Among all the sounds used by a given language, not all of them have *meaningful* differences
- ❑ Phonetics studies what is said; Phonology studies what is *meant*.
- ❑ **Phonemes** are a set of phonological units for a language, but a single phoneme means nothing by itself.

Vocal folds in action



COARTICULATION!



Q : Where can you see coarticulation ?

Faculté Polytechnique



Traitement de la parole

Modélisation



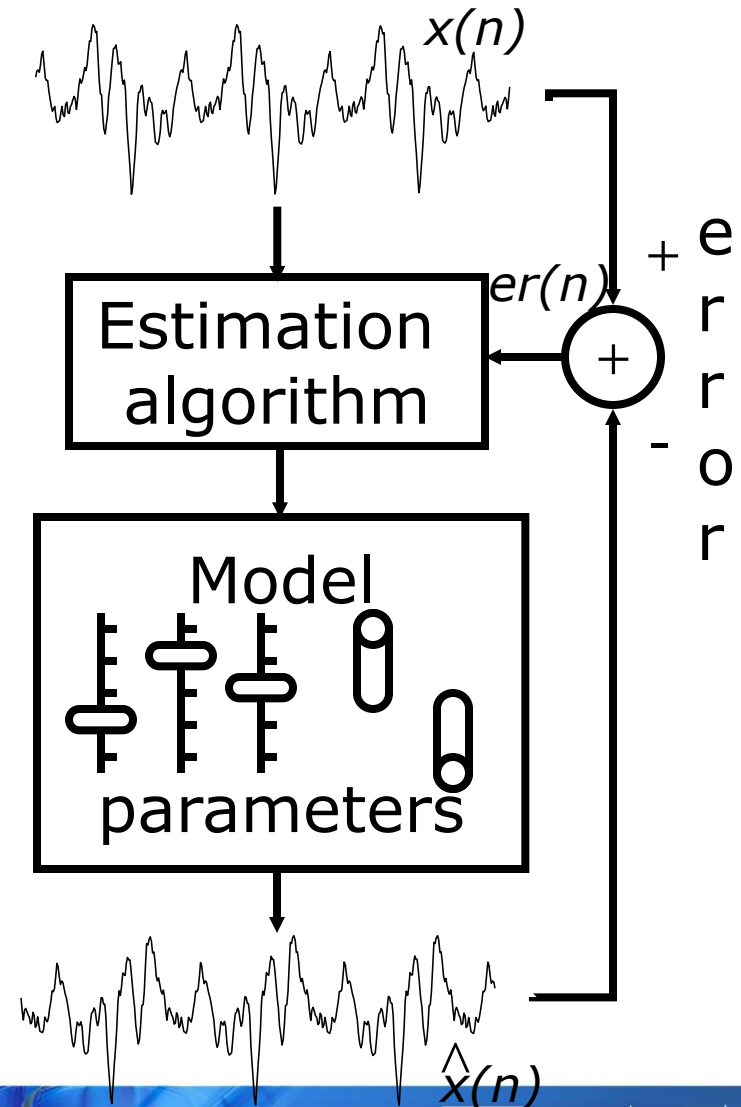
Carl Gunnar Michael Fant (1919, 2009) was professor at the [Royal Institute of Technology](#) (KTH) in [Stockholm](#). [Master of Science](#) in [Electrical Engineering](#), he specialized in the acoustics of the [human voice](#), measuring [formant](#) values, and continued to work in this area at [Ericsson](#) and at the [Massachusetts Institute of Technology](#). He also took the initiative of creating a speech communication department at KTH, unusual at the time.

Prof. T. Dutoit

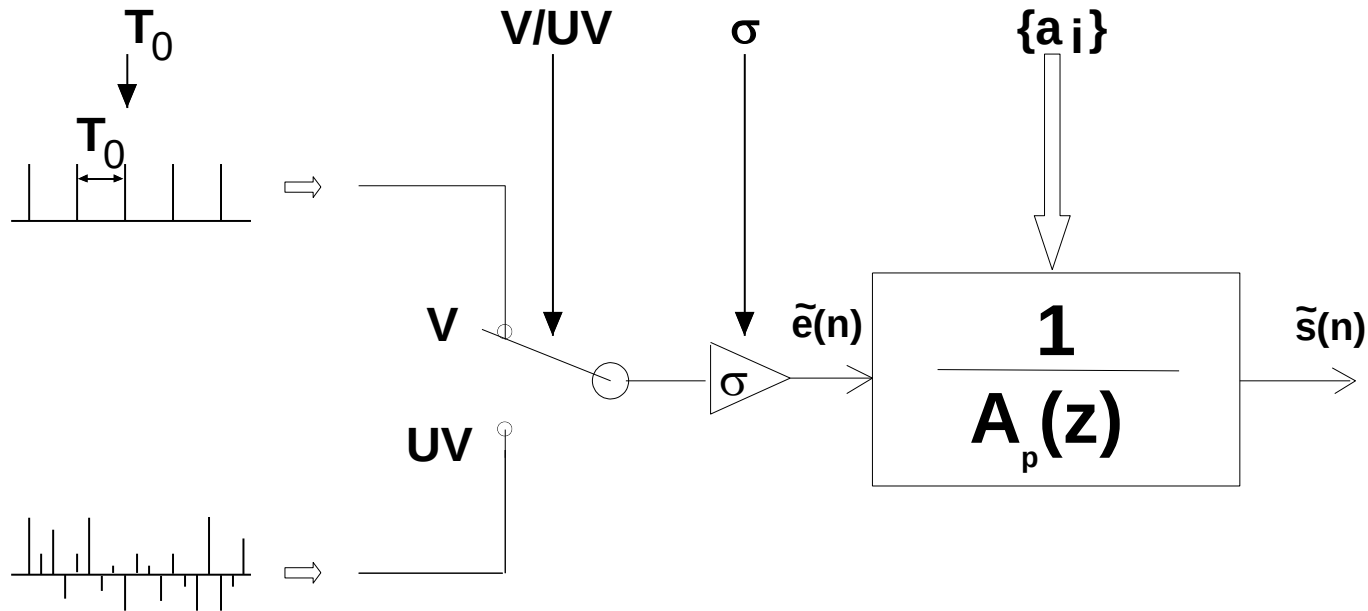
thierry.dutoit@umons.ac.be

Definitions

- ❑ Speech processing is based on *speech models*
- ❑ Models have *parameters*, like black boxes have switches and sliders
- ❑ Parameters are *estimated* via *algorithms*
- ❑ Errors : output of model $\neq$ input signal
 - **Modeling (intrinsic) errors**
 - **Estimation (extrinsic) errors**



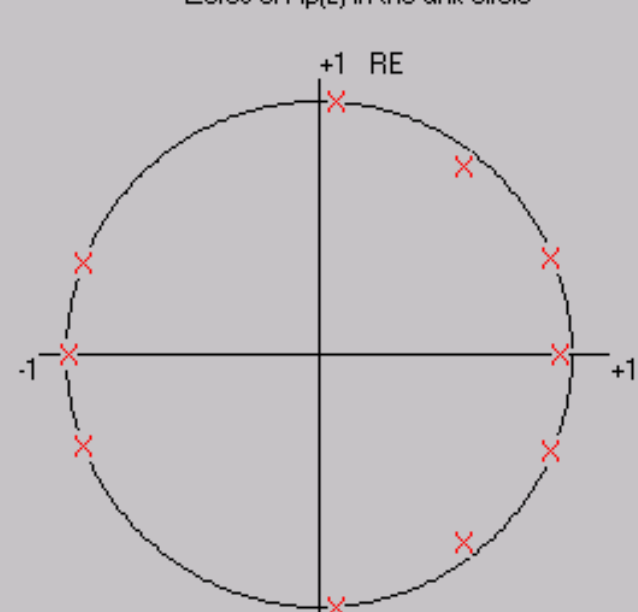
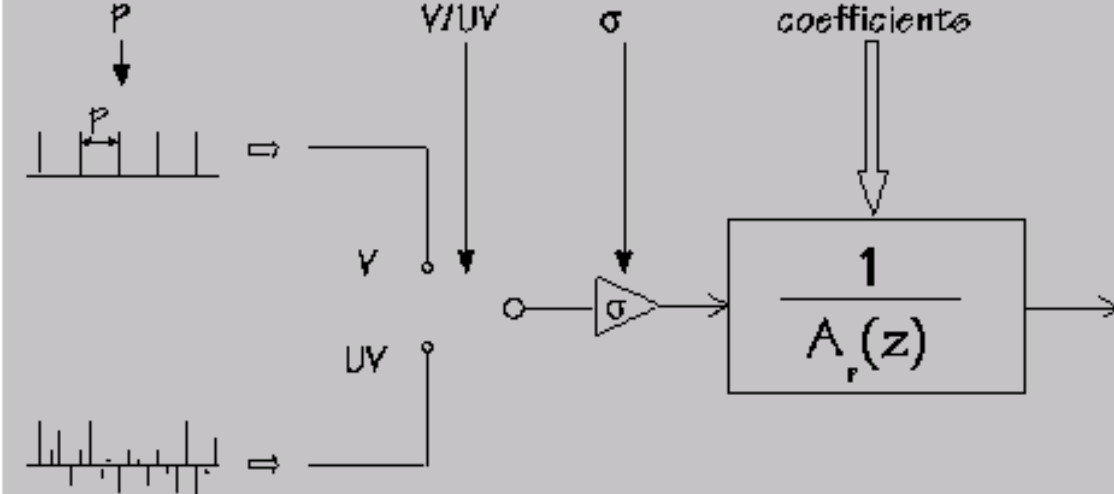
2. The LP model of speech production



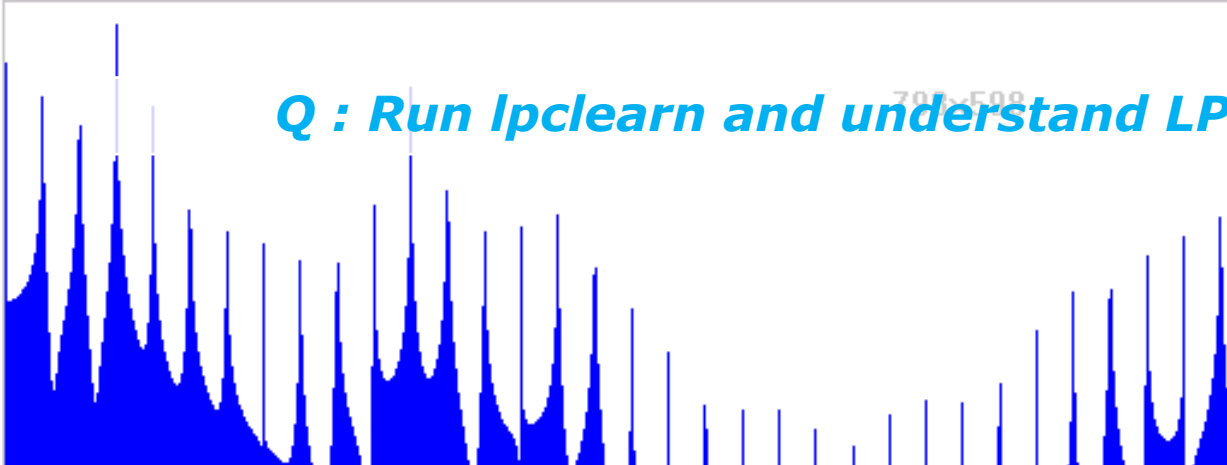
$$\tilde{S}(z) = \tilde{E}(z) \frac{1}{\sum_{i=0}^p a_i z^{-i}} = \tilde{E}(z) \frac{1}{A_p(z)} \quad (a_0 = 1)$$

$$\tilde{s}(n) = \tilde{e}(n) - \sum_{i=1}^p a_i \tilde{s}(n-i) \quad \Rightarrow \quad \text{LP} = \text{Linear Predictive} = \text{Autoregressive}$$

Q : How does the model produce voiced or unvoiced speech? How does it create formants?

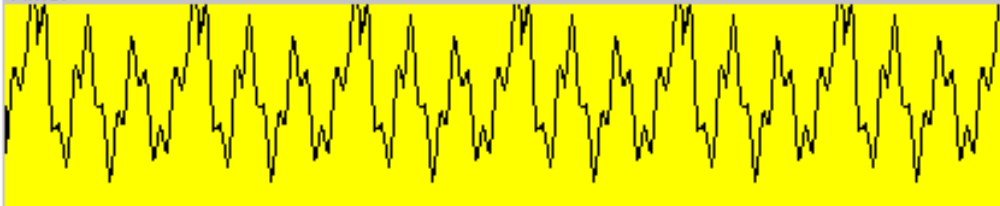


FFT:

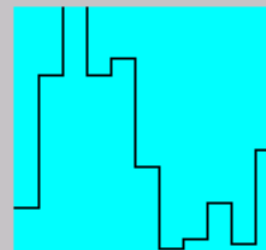


Q : Run lpclearn and understand LPC parameters

Time:



Tube model:



Pitch value: 118,52 (Hz)

PLAY REC Rec.gain: ☐ x2

The synthesized signal has to be: ☒ Voiced ☐ Unvoiced

Gain value: 1,00

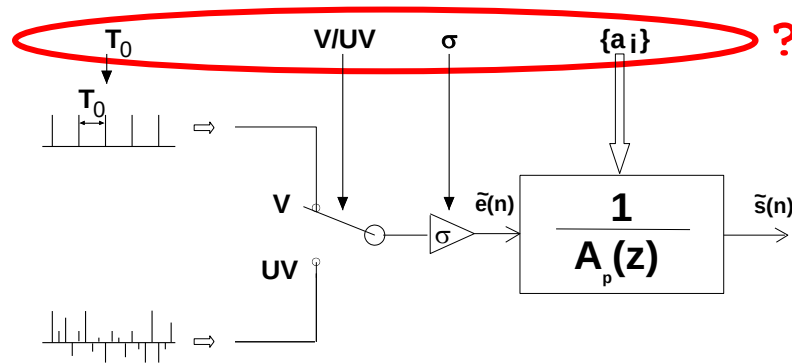
AGC

M+ MR

K1: -0,58	K2: -0,31
K3: 0,31	K4: -0,04
K5: 0,38	K6: 0,85
K7: -0,42	K8: -0,50
K9: 0,65	K10: -0,81

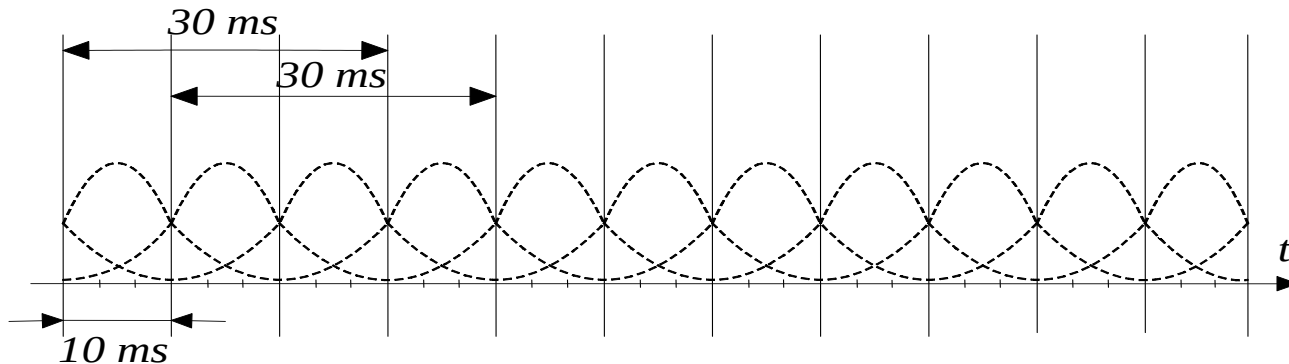
Ki Ai

3. LP estimation in 2 steps



1. T_0 and V/UV
 $\updownarrow$
independently
 σ and $\{a_0, a_1, \dots, a_p\}$

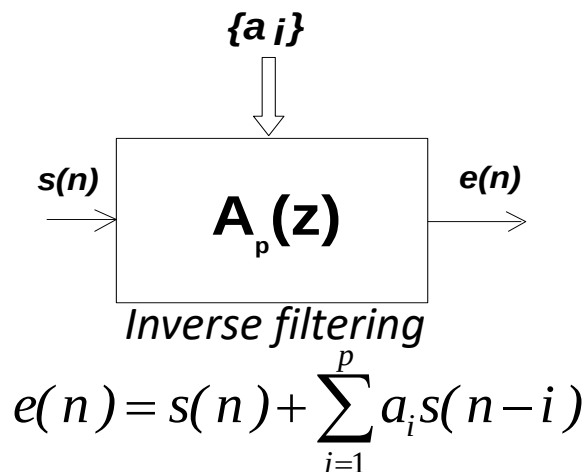
2. Speech is *pseudo-stationnary* on ≈ 30 ms



ex: $F_s=10\text{kHz}$, $L=100$ samples, $N=300$ samples

Q : What happens if the analysis frame is too large or too short?

LP estimation: σ and $\{a_0, a_1, \dots, a_p\}$



$$\{a_i\}^{opt} = \arg \min_{a_i} (E(e^2(n)))$$



Yule-Walker equations (linear)

$$\begin{bmatrix} \phi_{xx}(0) & \phi_{xx}(1) & \dots & \phi_{xx}(p-1) \\ \phi_{xx}(1) & \phi_{xx}(0) & \dots & \phi_{xx}(p-2) \\ \dots & \dots & \dots & \dots \\ \phi_{xx}(p-1) & \phi_{xx}(p-2) & \dots & \phi_{xx}(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_p \end{bmatrix} = - \begin{bmatrix} \phi_{xx}(1) \\ \phi_{xx}(2) \\ \dots \\ \phi_{xx}(p) \end{bmatrix}$$

$$\sigma^2 = \sum_{i=0}^p a_i \phi_{xx}(i)$$

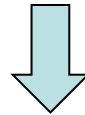
Q : How many operations to get the a_i coeffs from a 30 ms frame of speech?

LP estimation: σ and $\{a_0, a_1, \dots, a_p\}$

$$\begin{bmatrix} \phi_{xx}(0) & \phi_{xx}(1) & \dots & \phi_{xx}(p-1) \\ \phi_{xx}(1) & \phi_{xx}(0) & \dots & \phi_{xx}(p-2) \\ \dots & \dots & \dots & \dots \\ \phi_{xx}(p-1) & \phi_{xx}(p-2) & \dots & \phi_{xx}(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_p \end{bmatrix} = - \begin{bmatrix} \phi_{xx}(1) \\ \phi_{xx}(2) \\ \dots \\ \phi_{xx}(p) \end{bmatrix}$$

p linear equations
with p unknowns $\Rightarrow O(p^3)?$

NO! Toeplitz matrix $\Rightarrow O(p^2)$: recursively on p



LEVINSON algorithm

*Q : How many operations to get the a_i coeffs
from a 30 ms frame of speech?*

NB: p ?

- 2 poles/formant
- 1 formant/kHz
- $F_s/2$ kHz



$$p \approx F_s$$

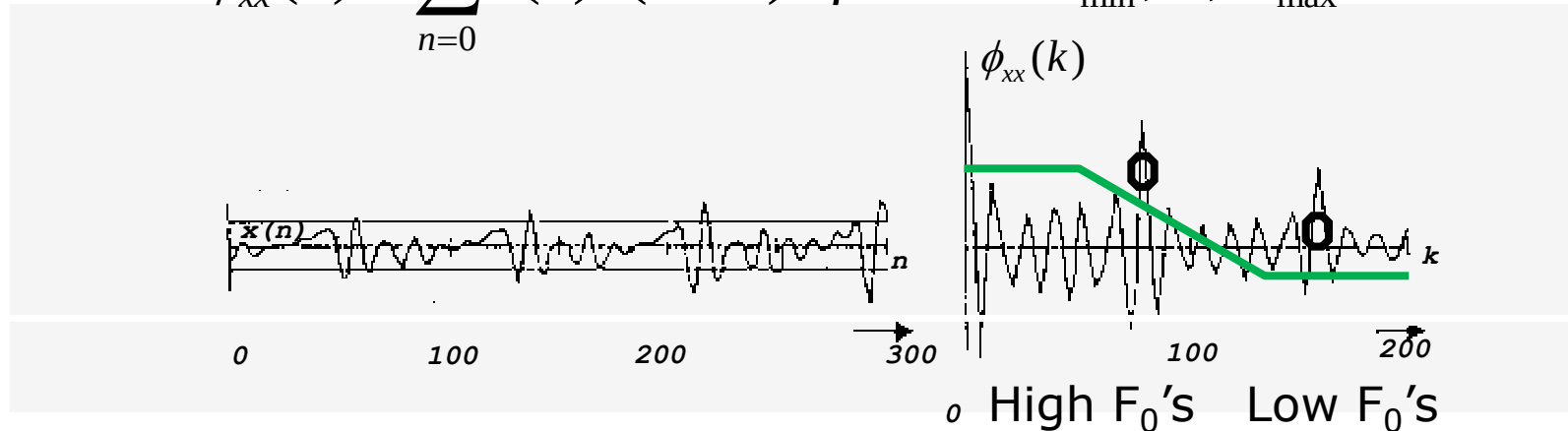
$$p \approx F_s + 2 \text{ (empirical)}$$

*Q Why « 2 poles per formant »?
Typical p for $F_s=8\text{kHz}$?*

LP estimation: T_0

Periodicity of x implies a max in $\phi_{xx}(k)$ for $kT_s \approx T_0$

$$\phi_{xx}(k) = \sum_{n=0}^{N-1} x(n)x(n+k) \quad \text{for } k = K_{\min}, \dots, K_{\max}$$



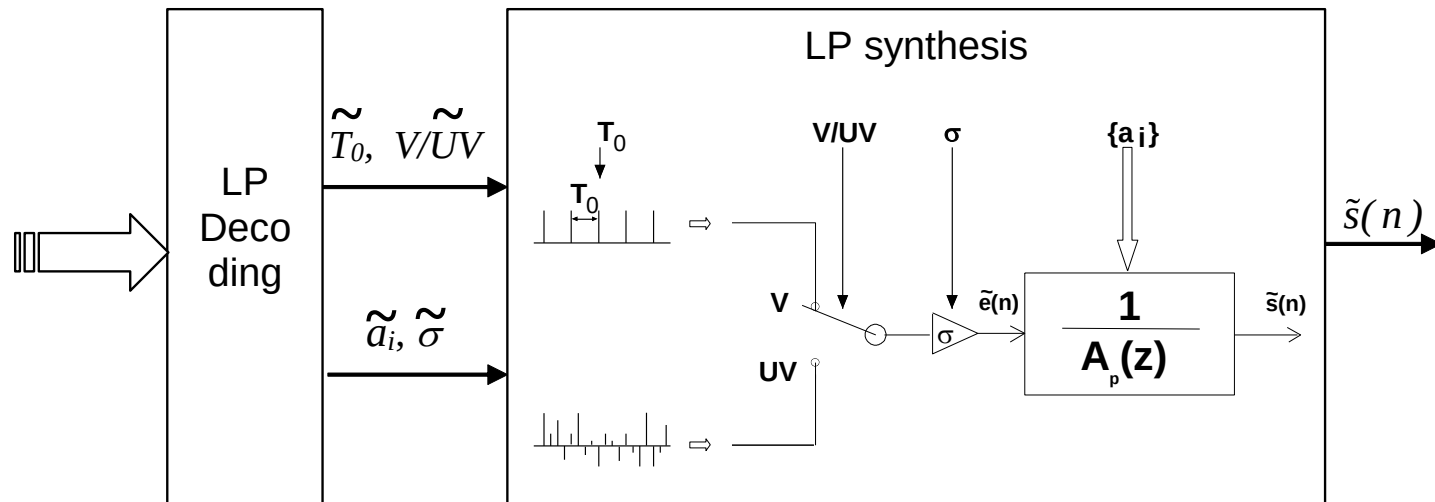
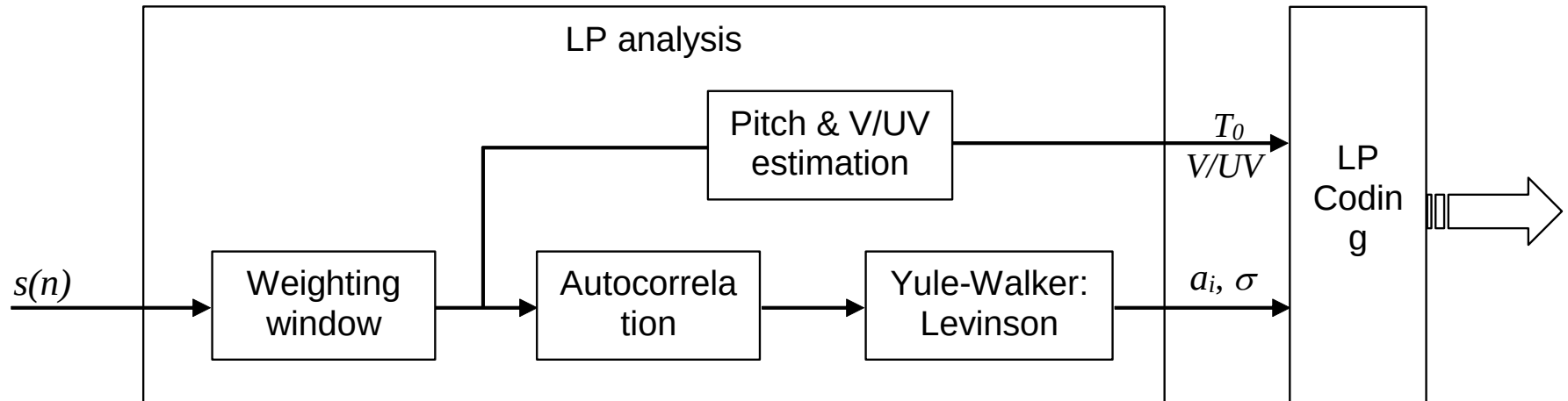
Computational load >>>

F0 doubling for females, because of $F1 \approx H2$

$\Rightarrow$ use **threshold(k)**

*Q : Estimate $K_{\min}$ and $K_{\max}$ for a male voice at
 $F_s = 8\text{kHz}$*

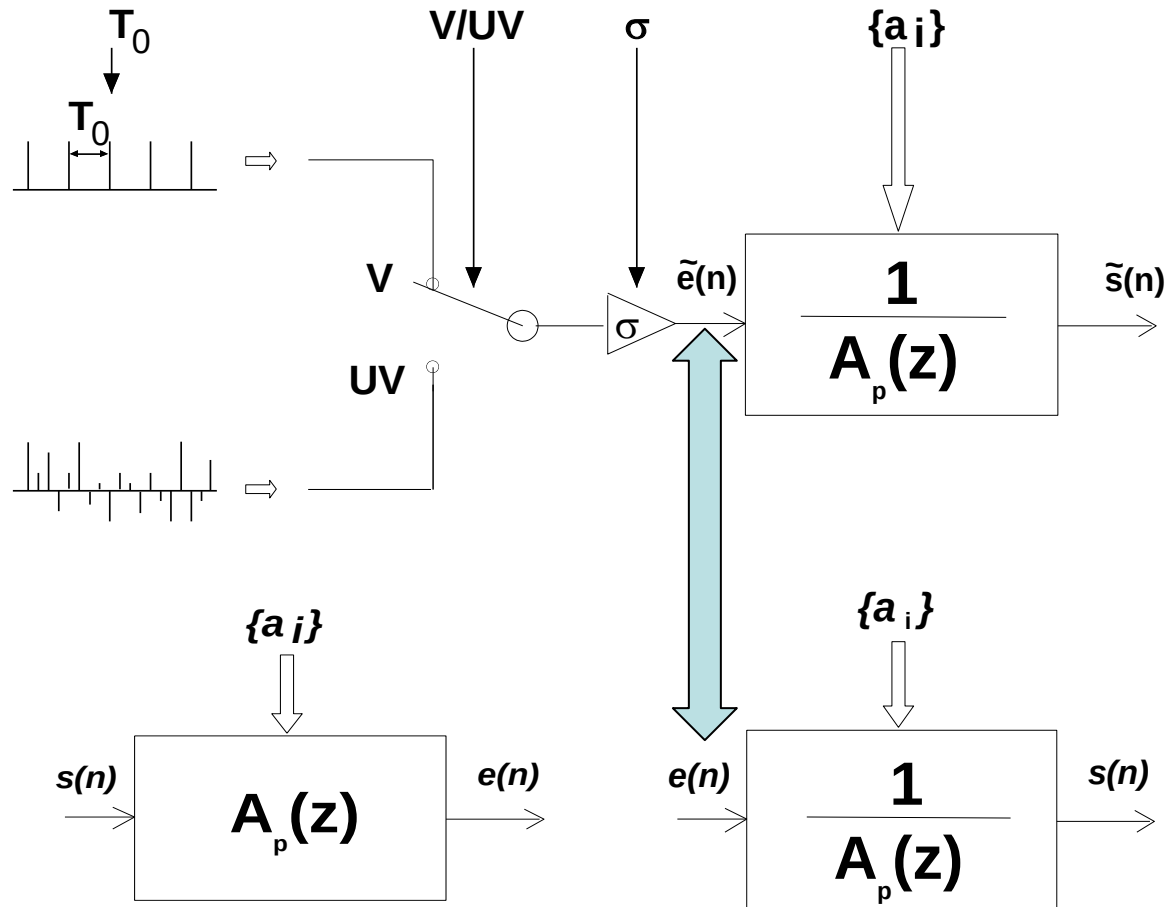
4. LP speech coding



Linear predictive coders: LPC10

- ❑ LP analysis/synthesis implemented in the so-called NATO LPC10 standard (NATO, 1984), used for satellite transmission of speech communications until 1996.
- ❑ This norm makes it possible to encode speech with a bit rate as low as 2400 bits/s
 - each 22.5ms-long frame is coded with 54 bits:
 - 7 bits for pitch and V/UV decision
 - 5 bits for the gain
 - 42 bits for the prediction coefficients

Enhanced LP coders



*Q : What is the inverse filter?
Why is $e(n)$ different from $\tilde{e}(n)$?*

Enhanced LP coders: MPE

(Atal and Remde 1982)

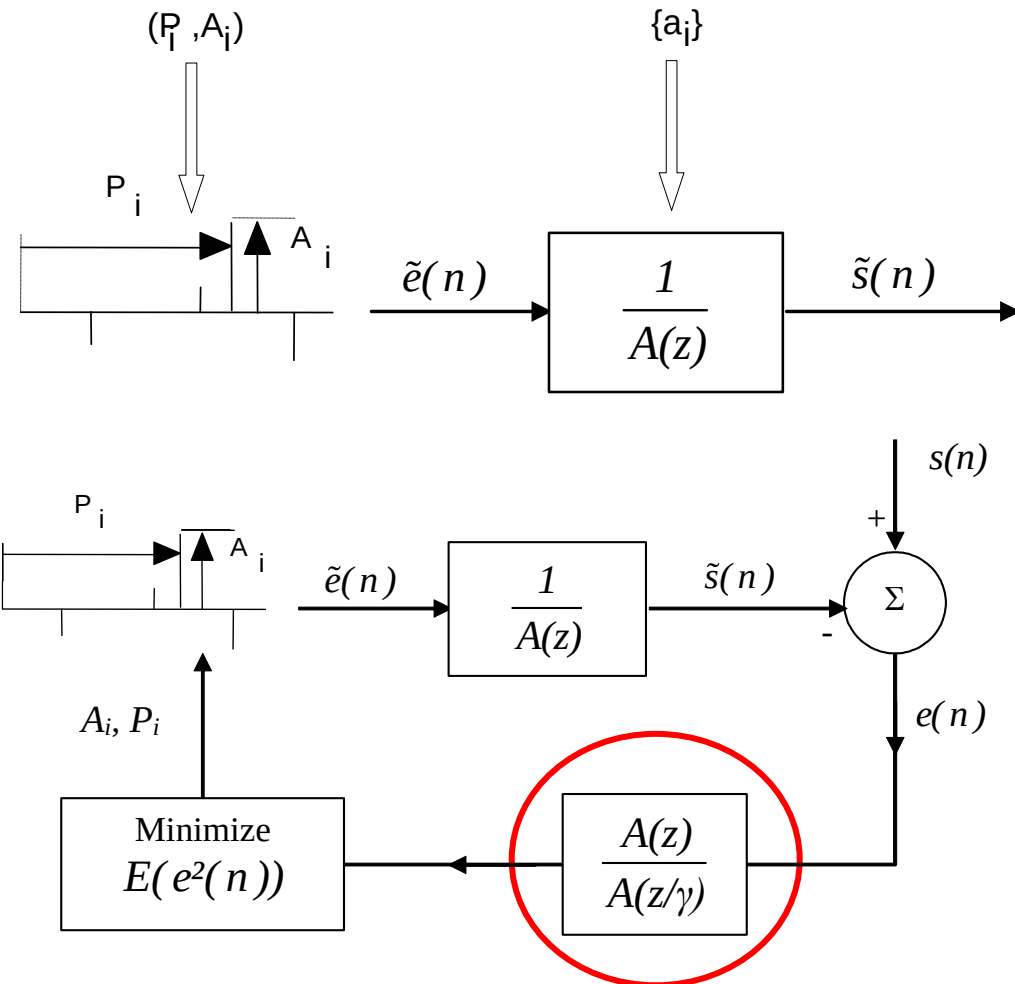
□ Positions and amplitudes of typically 10 pulses per 10ms frame

□ Analysis-**by**-synthesis process (i.e., a closed loop)

□ Error is filtered by a

perceptual filter

- $\gamma=1 \rightarrow$ All Pass (no effect)
- $\gamma=0 \rightarrow$ Inverse filter $A(z)$
- typically $\gamma=0.8$
- Accepts higher spectral components of $e(n)$ at formant positions

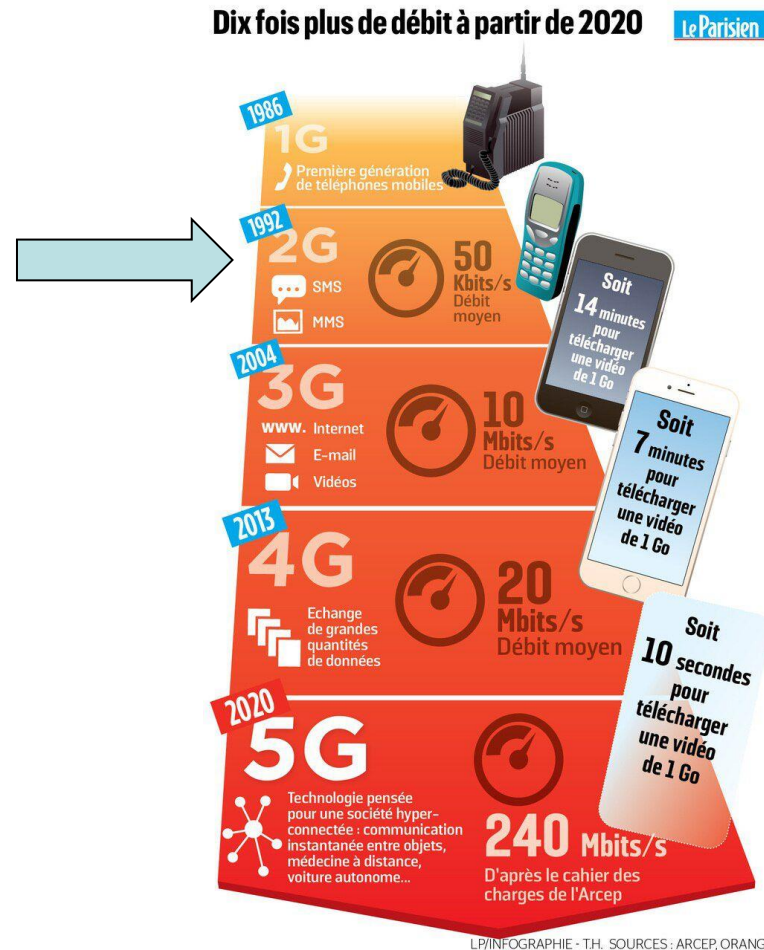


Q : How does the perceptual filter lead to higher spectral errors at formants?

A short history of the Cell Phone



Peter Kroon



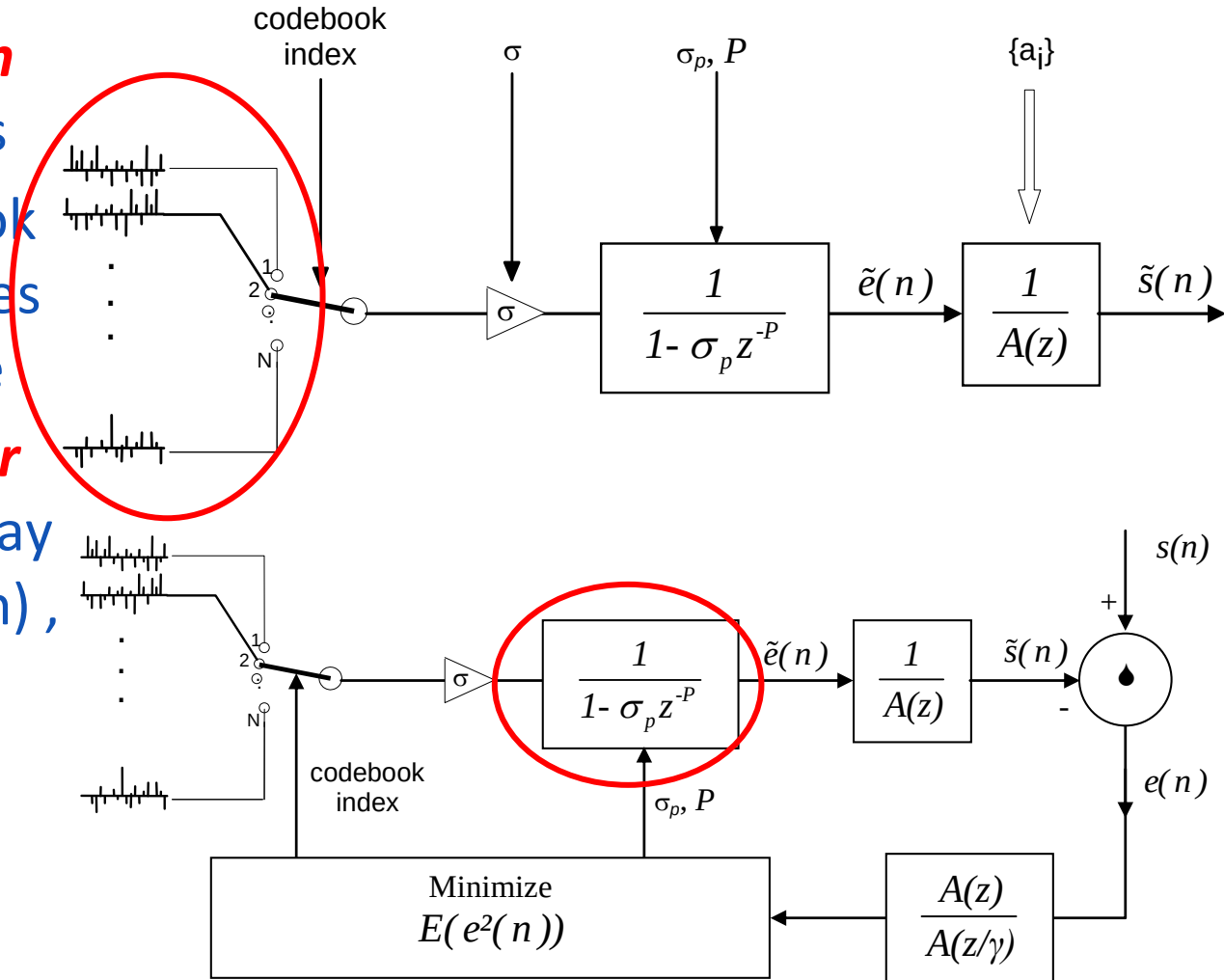
Enhanced LP coders: CELP

(Schroeder and Atal 1985)

❑ **Vector quantization**
of excitation frames

❑ Typ. 10-bit codebook
frames of 40 samples
→ 0.25 bits/sample

❑ **Long-term predictor**
filter (= variable delay
with adjustable gain),
enhances the
efficiency of the
codebook



Q : How does the long-term predictor produce periodicity?

Find Bishnu Atal 's bio.

LP coders in consumer products

- ❑ Enhancing CELP quality while further reducing bit-rate?
 - Mixed-Excitation Linear Prediction (**MELP**; McCree and Barnwell 1995). In 1996, LPC-10 was replaced by MELP to be the United States Federal Standard for coding at 2.4 kbps.
 - Harmonic and Vector eXcitation coding (**HVXC**; Matsumoto *et al* 1997).
- ❑ 1992-1996, GSM embedded the **RPE-LPC** (Regular Pulse Excited; Kroon *et. al* 1986) coder. Total bit rate of 13 kbps.
- ❑ 1996: Algebraic-CELP (**ACELP**, Salami *et al.* 1998) at 11.2 kb/s
- ❑ A variant of the ACELP coder has been standardized by ITU-T as **G.729** for operation at a bit rate of 8 kb/s. (used in VoIP)
- ❑ Newer generations of coders used in cell phones are all based on the CELP principle and can operate at bitrates as low as 4.75 kb/s to 11.2 kb/s.

